

Internet Technologies

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1 INTERNET FUELED INNOVATIONS

1.1 Categories

With the continued emergence of Internet fueled innovations, everyone in the world is constantly experiencing new technology every single day. The three categories of these innovations first involve transforming the physical world into the cyber world. Personally I have a very large amount of interaction with these types of innovations especially, since I work in data entry and record keeping. Even though today the world already has started to lean off of physical paperwork, there are still millions and millions of physical paperwork done every single day that do not get digitized. I have witnessed thousands of misplaced papers and files all throughout my time working in the government. With the continued implementation of online databases, the issues of physical paperwork are slowly being improved step by step. The second category is combining the physical world and cyber world. This can clearly be seen through the use of my glucose monitoring machine that I wear. Up until recently, if I wanted to get a blood sugar level monitoring I would need to take measurements every meal. However, with the implementation of this device, my doctor and I are able to constantly get my sugar level readings whenever and wherever we would like. This greatly bridges the physical and cyber world that this innovation was aiming for.



Figure 1—Continuous glucose monitoring machine. [Source](#)

The last category revolves around enriching the physical world. The best example that I can describe this is the new utilization of drones in the workforce. There have been many new services revolving around the use of drones in delivery, monitoring, and agriculture. Since drones can be outfitted with a large variety of devices, users are able to now use them for watering

plants, fixing machinery in high altitude, and even grocery shopping. These innovations have ultimately improved the lives of so many people already even though this use of drones has only recently started.

1.2 Commonalities/Distinctions

All three of these innovations are all very different in their own ways; however, the main thing that connects these innovations is that they are all used to help improve a problem that a user or society had as a whole. Databases have been around for a long time; however, it was not an enforced use in some departments in the government until recently when the issue of files and paperwork buildup has led to many problems. The database helped alleviate these issues and consolidate all the paperwork. My continuous glucose monitor helped with improving the issue of forgetting or not being able to take my blood sugar levels regularly. Finally the drones helped with many issues nowadays with online delivery, crop growing, and dangerous jobs. All of these were a great benefit to society over all. The commonality of helping society is something that all of these innovations have, but besides that there are many differences in these technologies. Each of them have different user communities. Databases have an overall flexibility that allows any person to use them for multiple reasons. However, drones and a glucose monitor are highly specific to specific targeted groups of people. This shows that innovations are highly varied and even though they are all designed to help society in some way they can be made for any kind of group out there.

1.3 Comparison

Looking over all the different types of Internet fueled innovations so far, society is constantly asking for innovations to be helpful to many people and as highly technologically advanced as possible. I completely agree that innovations should be helpful to as many people as possible; however, I believe that innovations geared towards helping smaller groups of people should also not be hindered. Society as a whole should be benefited, but smaller groups who have a need for innovations that can help in daily life are also important. I expect in the future that innovations can expand and use its process to improve other similar processes, such as drones being used in even more markets or the glucose monitor technology to be used in other illnesses as well.

2 WORLD WIDE WEB

2.1 Personal Activity

I am an avid user of the surface Web and the deep Web consistently every single day. Much like everyone else in today's digital age, I use the surface Web for mainly entertainment and school. For this example, I would like to use Google since I use this site the most. The deep Web on the other hand is something that I also use fairly often since I work for the government. Although I

am not able to specify the exact names of these databases, these tools that I use for work everyday are an example of the deep Web content since they are not indexed by search engines and cannot be found in simple queries. Both of these sites have many similarities and differences. The main thing that makes these sites so similar is that they have almost identical uses in many cases. Almost all of the functionalities that the government site has can be seen within Google as well. The only difference is that the government site has very heavy user authentication and monitoring that can restrict the user, while Google has much more flexibility in what it can do. In addition, the government site has access to more sensitive information. Besides these points, both sites are practically identical. These similarities are due to the sites' intended purpose. On the other hand, their differences are purely due to confidentiality of the information on the sites and user authentication.

2.2 Deep Web

Search engines today cannot search the deep Web ultimately due to the location of the data itself and the fact that the content is not indexed by the search engines. 96% - 99% of the content on the Internet nowadays are not indexed by the search engines, and are located in databases that typically require user authentication for access (Admin, 2023). The search engines, such as Google and Bing, are able to access the main webpage that is available on the surface Web. However, typically without user access, the deep Web portion of a site is inaccessible. To domain isn't typically able to be used to access content within the database. Once user authentication is completed then a dynamically generated domain can give access to deep Web content. A basic example of this is banking information. Anyone is able to access his or her bank's website which is located on the surface Web. However, any confidential user information is not able to be viewed without account user information. This authentication gives the user access to private user data that is located on the deep Web.

2.3 Deep Web Search Engine

Personally, I believe that the concept of a deep web search engine is not a viable project. There are many factors that contribute to this conclusion. The main issue that I come across is the protection behind a majority of the deep Web content. Personally I'm not too familiar with cyber security; however, the deep Web content is designed to be put behind paywalls, passwords, and privacy settings for a reason. To be able to consistently get past all of these defenses does not seem reasonable. Another issue I come across is the scalability of the project itself. The surface web accounts to approximately 10% of all the content on the Internet, while the deep Web covers 90% (Albano). This means that a search engine would have to be able to index nine times more than what it is currently indexing with the surface Web. This fact severely limits the scalability and speed of the search engine itself. The final issue that I notice will be a hurdle in a deep Web

search engine is the protection behind it. Although a majority of the deep Web is safe and perfectly legal, there can be dangerous parts that can still lead people towards cyber threats and danger. Ethically I do not believe giving the public easy access to potential harm is something that should be advocated with a deep Web search engine.

3 WEB CRAWLER

3.1 PULSE

Steve Proell's PULSE crawler was ultimately an extremely inefficient way for crawling the web due to the lack of case handling within the crawler itself. Based on the crawler's design, it was able to specify the length limit when locating subjects and the search strategy. While this is decent, it does not give consideration to edge cases and loops that may occur. In addition, it did not give flexibility in its restriction of how the crawling is done. Key words could not be searched or filtered out. Ultimately, there are numerous ways that the speed and accuracy of the web crawler could have been improved.

3.2 Georgia Tech Crawler

If I was asked to design a crawler to crawl the Georgia Tech domain, the main thing I would focus on is the speed and flexibility of the web crawler. Since the crawler will be designed for a single domain, the efficiency of the crawler is especially important. I would add in functionality to be able to search or filter out specific keywords that can speed up the crawling of the domain. The key words can allow the crawler to partially read pages instead of reading the entire page or up to a certain point. This can somewhat cut down on time. Ultimately, I believe a goal of 10,000 pages per hour would be good. Now the database holding the subjects would still result in holding an enormous amount of information. To restrict this, all the individual domains will be put into a hashtable to eliminate duplicates. Another thing to add to the crawler to help with the storage cost is the improvement of the crawling of each page. When the HTML page is retrieved and scanned, the text of the HTML page can be taken and the Boyer-Moore string search algorithm can be used to speed up the searching of keywords in the text instead of reading the entire text. All of this would be what I would implement in my web crawler of the Georgia Tech domain.

3.3 Crawl the Web

Although my web crawler would definitely be more efficient than the PULSE crawler, it would still be highly inefficient in web crawlers. If the target crawling speed is 1 billion pages per year, and my web crawler's goal of crawling the Georgia Tech domain was 10,000 pages per hour then the estimate of my crawler would be approximately 87.6 million pages per year. This is targeted pages per hour for a web crawler would be 114155 pages per hour.

4 SEARCH ENGINE RANKING ALGORITHMS

4.1 Importance

Ranking algorithms are important for a search engine because it is the factor that gives relevance to all the pages in the index. Without this relevance, you would be receiving a multitude of pages that range in topics to what you are looking for. Keywords, frequency, and popularity are all taken into account to be able to determine the relevance of each page to be ranked. The relevance is ultimately what determines what pages are seen first to best optimize usability and efficiency. Without the ranking, the user will end up seeing way too many pages in a random order that will severely cripple usability. The pages the user will be looking at can end up having little to no information regarding what is needed. This will in turn force the user to look at more pages and ultimately reduce efficiency. The ranking algorithms' importance can greatly affect the usability and efficiency of the search engine.

4.2 Ranking Method Comparison

	Query Dependent	Query Independent
Advantages	Runs web pages according to the quality of the link and the quality of the content Good source of links and content	Run according to global ranking value Important page is linked by many pages or by other important pages
Disadvantages	Dependent on the return of the query and ranking is calculated based on only the return from the query	Ranking is done regardless of the actual query from the global pages

4.3 Google PageRank

Google PageRank has its many uses and can be very helpful in many instances. However, there can also be times where it does not give much assistance in the tasks I am doing. Depending on the context of why I am using Google's search engine, the PageRank can both help and hurt the process of trying to find content. For example, the location I live in heavily relies on Google to search for popular restaurants and food locations instead of Yelp or other similar apps. This results in the search engine returning the most popular and mainstream restaurants in my area based on the webpage's popularity. Although it can more often than not give me the more favorable restaurants in the area, this does not mean that the food at the restaurant is equivalent

to the popularity the restaurant has on the web. Another similar instance of this is when a personal friend was able to manipulate the traffic of his shop's website to increase his site's PageRank. He owned a smaller lesser known shop that was being overshadowed by big brand name shops. However, eventually he was able to artificially manipulate the popularity and traffic of his site to increase the Google PageRank of his site. This just proves the disparity and inaccuracies that can stem from Google's Page Rank. The PageRank is definitely useful to give quick access to the most popular pages based on the keywords; however, these pages cannot always accurately give the proper sites.

5 REFERENCES

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